



ai●palette

ROI LENS

GitHub Link:

<https://github.com/Ashutosh-singh-data/ROI-LENS>



Summer Project

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ROI Lens: Advanced Marketing Attribution & Budget Refinement

Q1 2026 | Nexus Consumer Brands | ₹100 Crore Campaign Audit

The Business Problem

Nexus deployed ₹100 Crore across 5 channels:

- Instagram | Google Search | YouTube
- Influencer Blog | E-commerce Marketplace

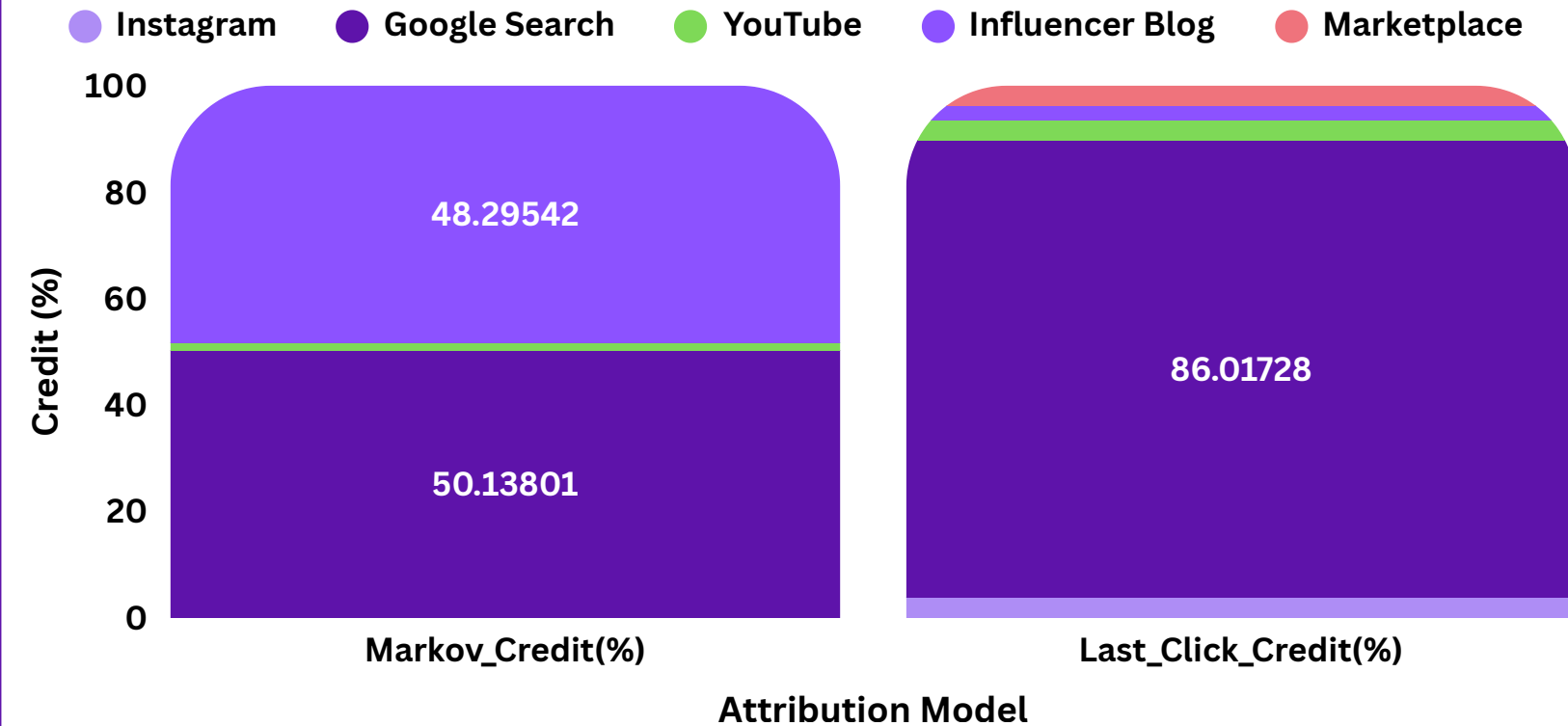
Across 10 brands, 100,000 customers

Current Model: Last-Click Attribution



Budget decisions based on flawed data

B02 Last-Click vs Markov

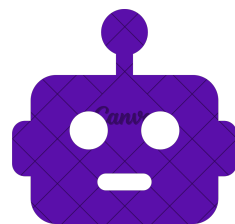


Key Findings

- YouTube received ₹26.6 Cr budget
→ Near-zero attribution across all brands
→ ₹25 Crore effectively wasted
- Instagram (B01) CPA = ₹964
→ Had only ₹61L budget
→ Was driving 92% of B01 conversions
- YouTube over-credited across entire portfolio
→ Last-Click gave it 12% credit average across all brands.
→ True Markov attribution: near zero across all brands.
- B07 — 100% single channel dependency
→ Strategic risk: zero fallback

The ROI LENS Solution

Data Integrity & Bot Elimination



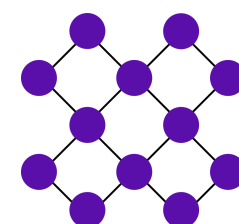
Bot Detection by 3 Behavioral Rules

Rule - 1: Speed Anomaly — events under 5 seconds apart (3+ occurrences).

Rule-2: Click Fraud — click rate above 50% of impressions.

Rule - 3: Campaign Spam — same campaign hit 15+ times.

Markov Chain Multi-Touch Attribution



Probabilistic state-based attribution model

- Every customer journey mapped as a sequence of states.
- Transition probability matrix built per brand (10 matrices total).
- States: Start → Channel interactions → Conversion/Dropout.
- Removal Effect applied: each channel removed one at a time.
- Conversion drop measured = true channel contribution.
- Negative effects clipped to zero, results normalized to 100%.

CPA-Driven Budget Reallocation



CPA-Driven Optimization Model

- Efficiency Score = $1 / \text{CPA}$ (lower CPA = higher score).
- Budget weight assigned proportional to efficiency.
- Diminishing returns cap: no channel exceeds 60% of brand budget.
- Iterative capping ensures total always equals ₹10 Crore exactly.
- Channels with zero attribution credit receive zero budget.

Dataset Overview: 3-Source Data Ecosystem
Q1 2026 | Nexus Consumer Brands | 10 Brands | 5 Channels

Data Volume

- 566,510 raw touchpoints recorded across Q1 2026.
- Represents every single customer interaction from first ad impression to final purchase.
- 100,000 unique users tracked across 10 FMCG brands.
- Dataset simulates a real-world marketing ecosystem with intentional noise — bot traffic, ad fatigue, and non-linear conversion paths included.

Consumer Intelligence

- 100,000 user profiles with demographic and psychographic attributes.
- 4 distinct consumer personas identified:
 - Fitness Enthusiast | Gen-Z Trendseeker
 - Premium Gourmet | Budget Parent
- 3 geographic tiers covered (Tier 1, 2, 3).
- Each user linked to trend affinities:
 - Vegan | Sustainable Packaging | Value-Pack
 - Luxury | High-Protein

Data Quality Challenges

- Bot traffic artificially inflated impression counts.
- 3,155 fake users detected — 3.2% of total traffic.
- Bots evenly distributed across all 5 channels indicating systematic injection not random noise.
- Diminishing returns present — repeated ad exposure yields lower engagement over time.
- Direct proportional analysis would yield mathematically flawed business advice.

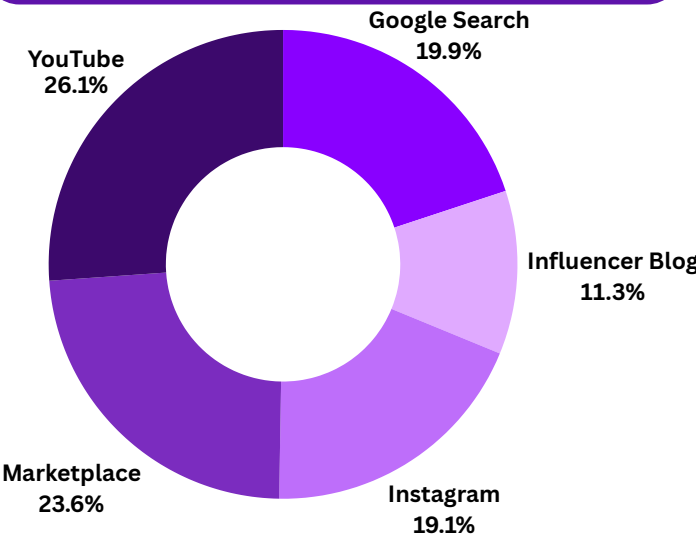
Channel Architecture

- 5 advertising channels tracked simultaneously: Instagram, Google Search, YouTube, Influencer Blog, E-commerce Marketplace.
- Each channel uses different pricing model: Google Search = CPC (pay per click) All others = CPM (pay per thousand impressions).
- Cannot be compared directly — normalization required.
- Total budget deployed: ₹101.86 Crore across 50 campaigns

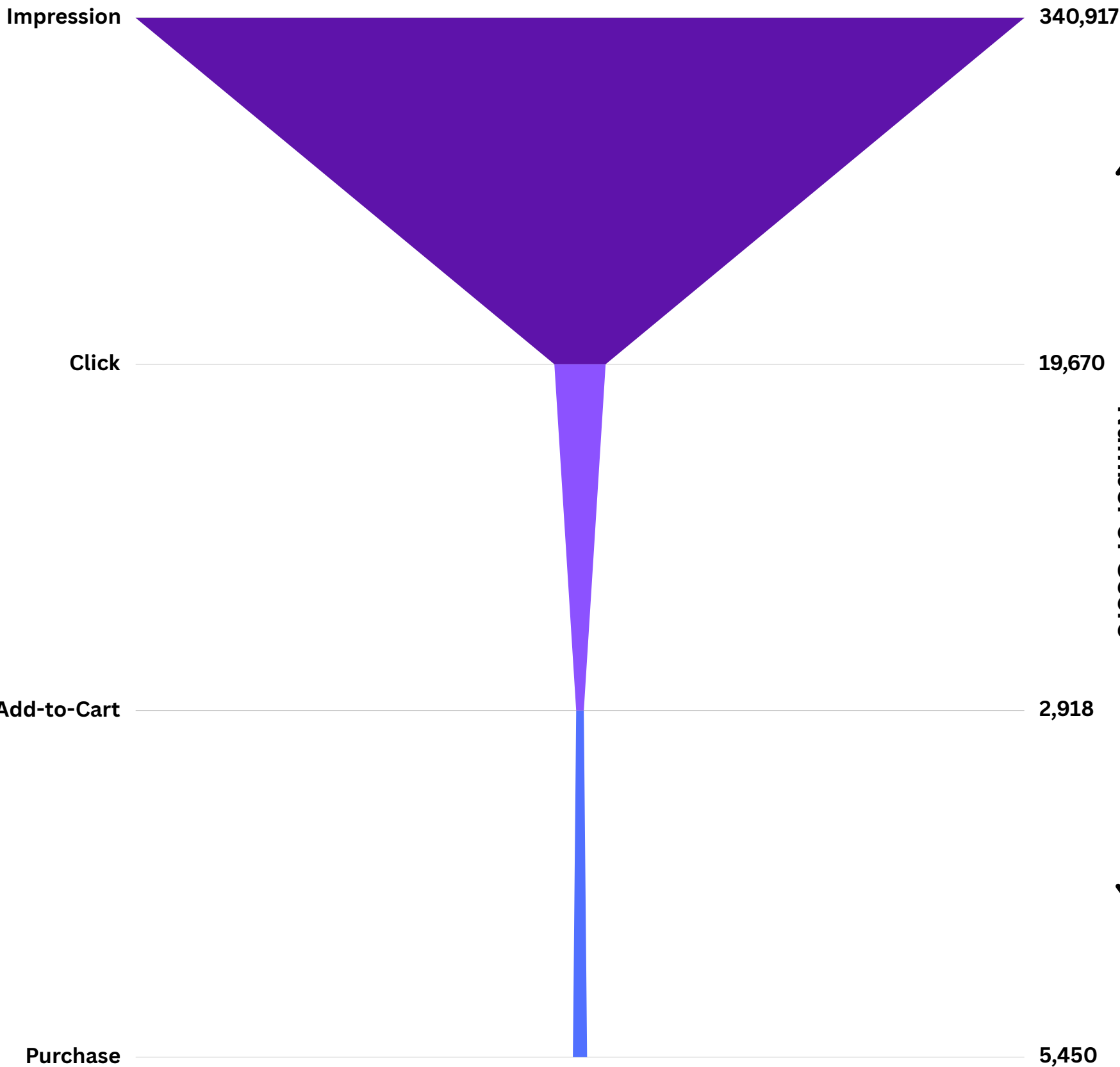
Conversion Funnel

- 4 event types tracked in sequence:
 - Impression → Click → Add-to-Cart → Purchase
- Only 5,450 users (5.6%) reached final purchase.
- 94.4% dropped off before converting.
- Average journey length: 5.8 touchpoints.
- Non-linear paths common — same channel appears multiple times in one journey.

Channel Budget Distribution



Conversion Funnel



Data Quality Audit: Eliminating Bot Traffic Before Attribution

3,155 Fake Users Detected | 3.2% of Total Traffic | Systematic Injection Confirmed

Why Bot Matter?

Raw data cannot be trusted directly. The dataset contains intentional real-world friction — bot users designed to mimic human behavior but with statistically impossible patterns.

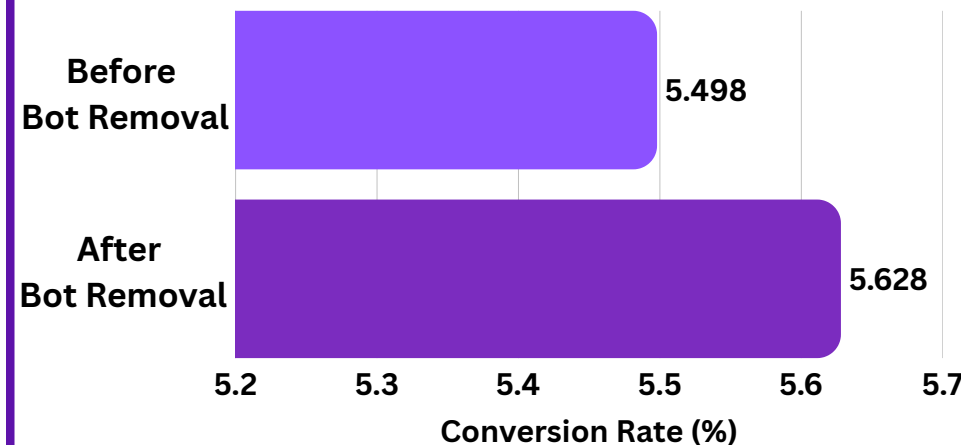
If bots are not removed before attribution:

- Channel credit gets artificially inflated.
- CPA calculations become mathematically invalid.
- Budget reallocation based on corrupted signals.
- CMO makes ₹100 Crore decisions on fake data.

Industry Reality

- Over 50% of web traffic is non-human.
- Digital advertisers lost \$84 Billion to ad fraud in 2024.
- Bot injection is systematic — not accidental.
- Even sophisticated platforms miss bot traffic without behavioral anomaly detection.

Conversion Rate (%)



3 Detection Rules

RULE 1 — SPEED ANOMALY

Detection

Events under 5 seconds apart occurring 3+ times per user.

Logic

Real humans cannot process an ad, decide to click, and complete checkout in under 5 seconds. Bots operate at machine speed.

Bots caught: 1,965 users

RULE 2 — CLICK FRAUD

Detection

Click rate exceeding 50% of total impressions seen.

Logic

Real humans click 1-5% of ads they see. A 50%+ click rate is statistically impossible for genuine consumer behavior.

Bots caught: 3,151 users

RULE 3 — CAMPAIGN SPAM

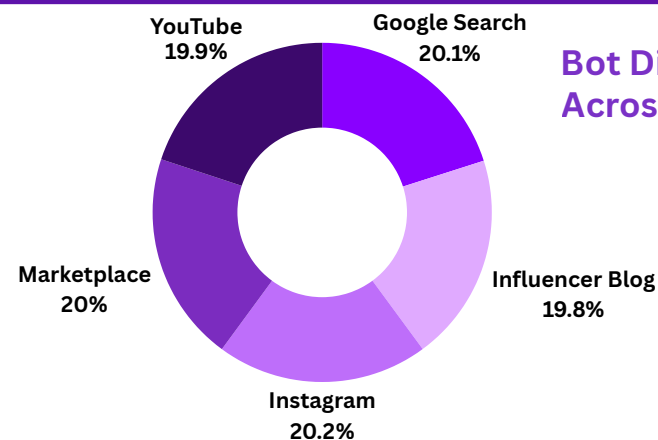
Detection

Same campaign hit 15+ times by single user.

Logic

Real users don't repeatedly engage with identical ad content. Repetitive hits signal automated bot behavior loops.

Bots caught: 1,942 users



The Critical Finding

This is NOT random. Equal distribution across all 5 channels confirms **SYSTEMATIC** bot injection — not organic fraud from one channel.

Implication:

Every channel's performance metrics were equally corrupted. Last-Click attribution was running on universally compromised data.

Cleaning Results

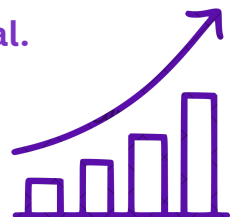
Before Cleaning

Total users: 100000
Conversion Rate Before Cleaning: 5.498%
Bot Users: 3155

After Cleaning

Clean users: 96845
Conversion Rate After Cleaning: 5.628%
Bot Removed: 3155 (3.2%)

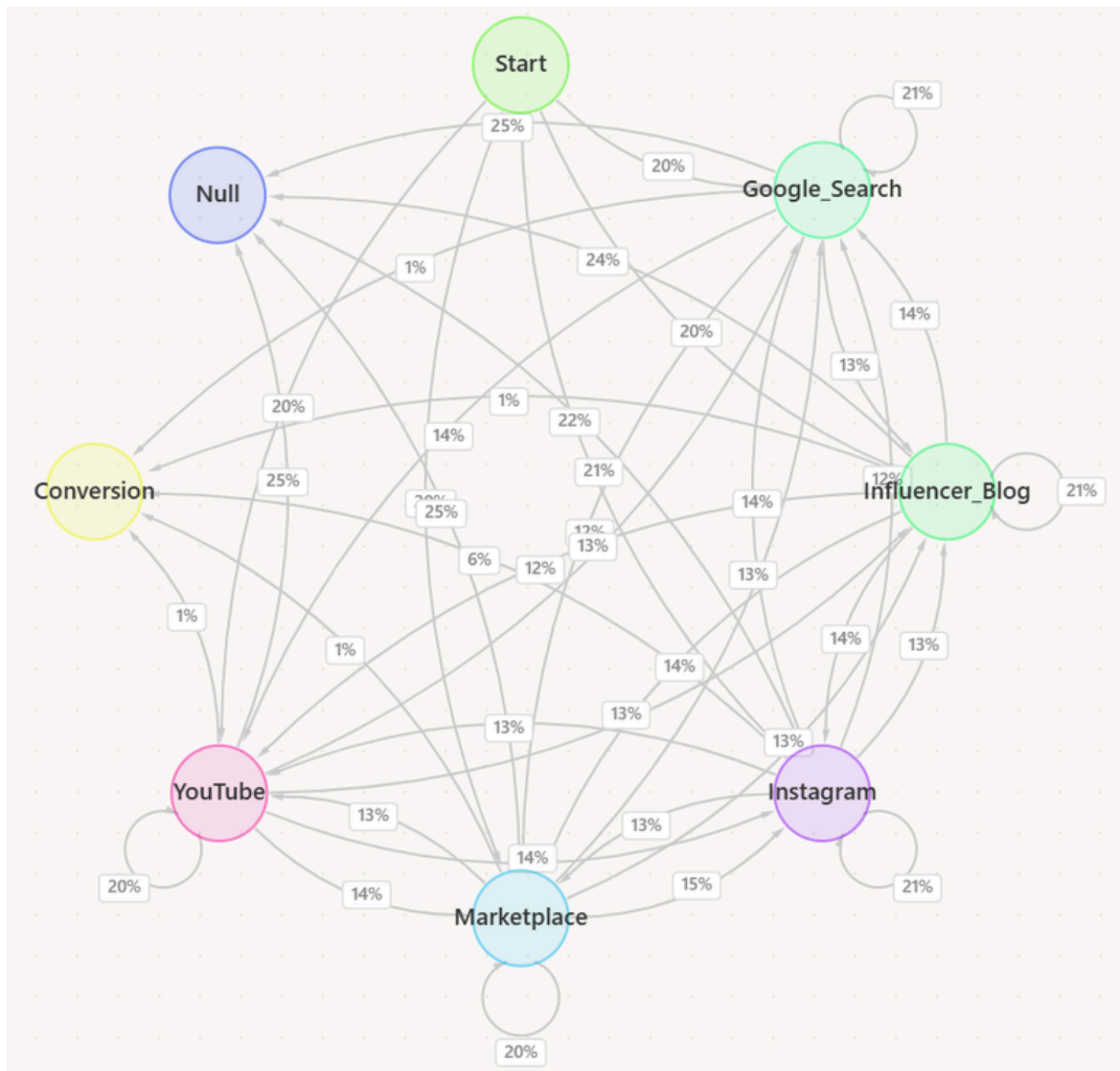
Conversion rate increased after cleaning — bots were suppressing true conversion signal.



Markov Chain Multi-Touch Attribution: How We Found The Truth

Why Last-Click Fails | How Markov Works | What Removal Effect Proves

Markov Chain Model for B01



We use B01 data is purely to illustrate the Markov Chain methodology.

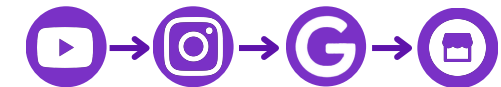
Transition Matrix for B01

to_state	Conversion	Google Search	Influencer Blog	Instagram	Marketplace	Null	YouTube
from_state							
Google Search	0.007449	0.211809	0.131771	0.142064	0.124594	0.245395	0.136918
Influencer Blog	0.008296	0.135999	0.214198	0.137903	0.136815	0.243574	0.123215
Instagram	0.060287	0.120446	0.127886	0.209082	0.128143	0.220498	0.133658
Marketplace	0.007019	0.132122	0.132810	0.148225	0.195706	0.249931	0.134187
Start	0.000000	0.202712	0.195155	0.206233	0.199814	0.000000	0.196087
YouTube	0.007256	0.131846	0.131161	0.141566	0.135405	0.250685	0.202081

Why Last-Click Fails

Last-Click attribution operates on a fatal assumption:
"The channel where purchase happened deserves 100% of the credit."

Real Customer Journey.



Last-Click gives Credit

0% 0% 0% 100%

What this causes

- YouTube budget gets cut (looks useless).
- Instagram starved of investment.
- Marketplace over-funded.
- Top-of-funnel channels die slowly.
- Eventually Marketplace sales drop too because nobody enters the funnel.

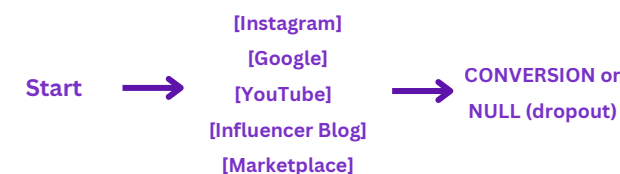
How Markov Works

How we build it

Core Principle:

Customer journey = sequence of states.
Each state transition has a probability.

States in our model:



Step 1

Extract every consecutive channel pair from all journeys.

Step 2

Count all transitions.

Step 3

Convert to probabilities
 $P(\text{YouTube} \rightarrow \text{Google}) = \text{count}(Y \rightarrow G) / \text{all YouTube transitions.}$

Step 4

Build 8x8 matrix per brand.

Step 5

Calculate conversion probability from Start to Conversion state.

Step 6

Apply Removal Effect.

Removal Effect: The Key Insight

Definition:

Remove one channel completely.
Recalculate conversion probability.
Measure the drop.

$$\text{Removal Effect} = (\text{Baseline Conv\%} - \text{New Conv\%}) / \text{Baseline Conv\%}$$

Example – Brand B01:

With all channels: 7.14% conversion

Remove Instagram: 3.25% conversion
Instagram Effect = $(7.14 - 3.25) / 7.14 = 54.5\%$

Remove YouTube: 7.13% conversion
YouTube Effect = $(7.14 - 7.13) / 7.14 = 0.13\%$

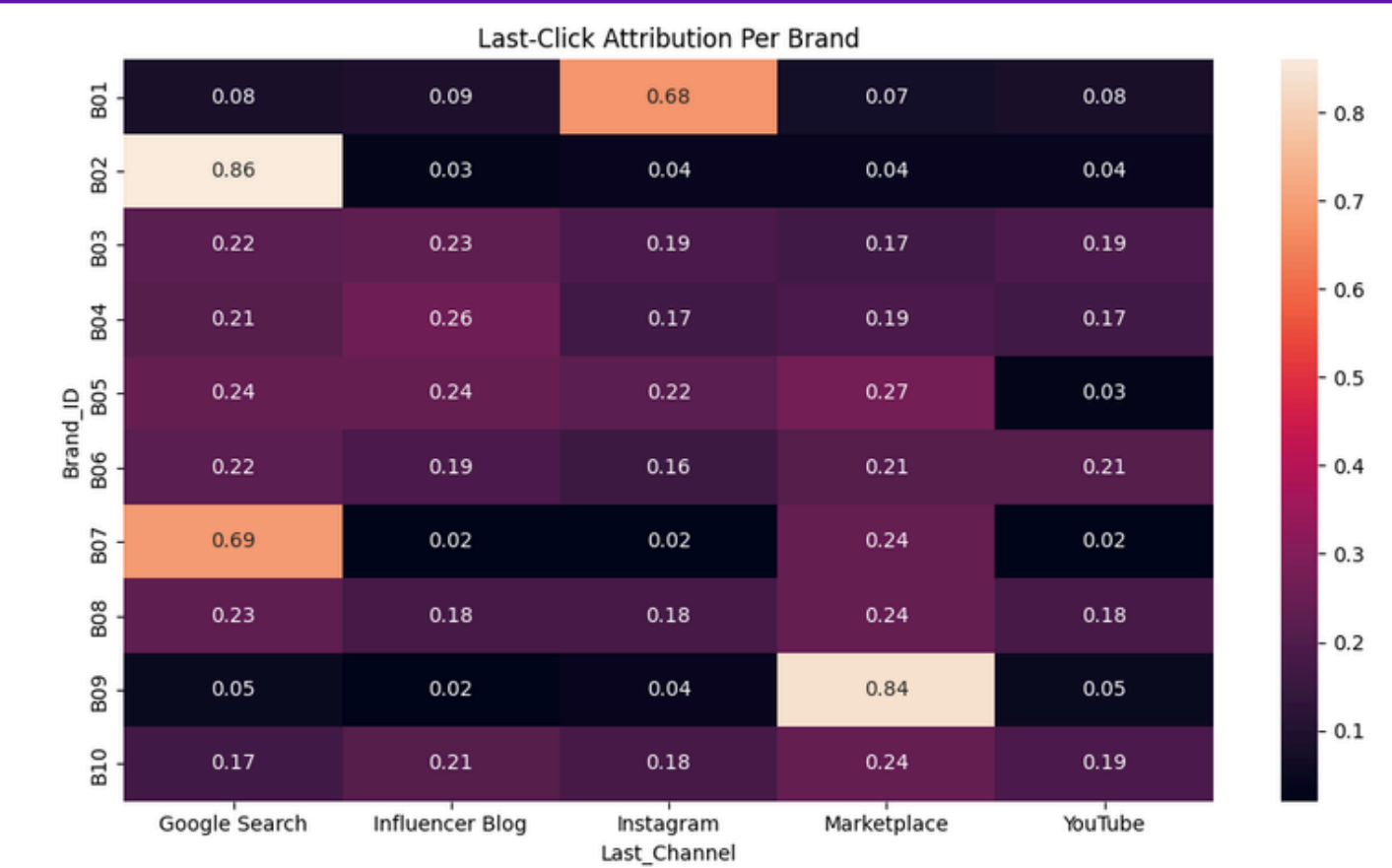
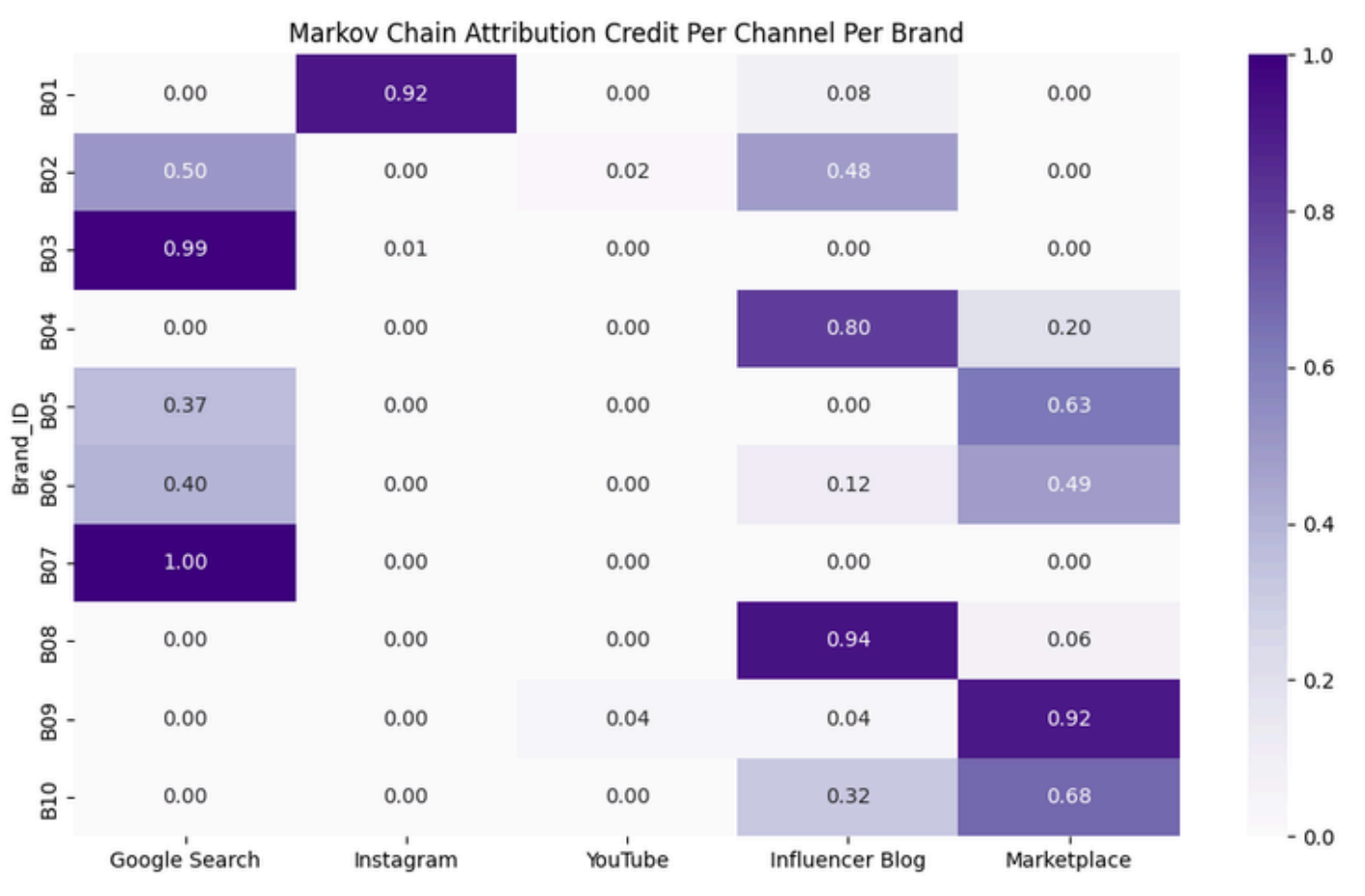
Conclusion:

Instagram drives 54.5% of B01 conversions
YouTube drives only 0.13%.

Negative effects → clipped to zero
All effects → normalized to 100%.

Attribution Results: Last-Click vs Markov Chain

What Nexus Believed vs What The Data Actually Shows



Key Differences Table

Brand	Last-Click Says	Markov Says	Biggest Shift
B01	Instagram 68%	Instagram 92%	Instagram +24%
B02	Google Search 86%	Google 50% + Influencer 48%	Influencer +45%
B03	Fairly equal (~22% each)	Google Search 99%	Google +77%
B04	Fairly equal (~20% each)	Influencer Blog 80%	Influencer +60%
B05	Marketplace 27% leads	Google 37% + Marketplace 63%	Marketplace +36%
B06	Fairly equal (~20% each)	Google 40% + Marketplace 49%	Marketplace +28%
B07	Google Search 69%	Google Search 100%	Google +31%
B08	Marketplace 24% leads	Influencer Blog 94%	Influencer +76%
B09	Marketplace 84%	Marketplace 92%	Marketplace +8%
B10	Marketplace 24% leads	Influencer 32% + Marketplace 68%	Marketplace +44%

Critical Findings

- 1 Last-Click distributes credit evenly (~20% per channel) for 8/10 brands.

→ Markov reveals each brand has ONE dominant channel.

→ Last-Click was masking this completely.
- 2 YouTube gets near-zero Markov credit across all 10 brands.

→ Last-Click was giving it ~12% credit everywhere.

→ ₹26.6 Crore budget allocated based on false signal.
- 3 Influencer Blog is the hidden hero

→ Last-Click average credit: ~15%.

→ Markov average credit: ~40% across brands where active.

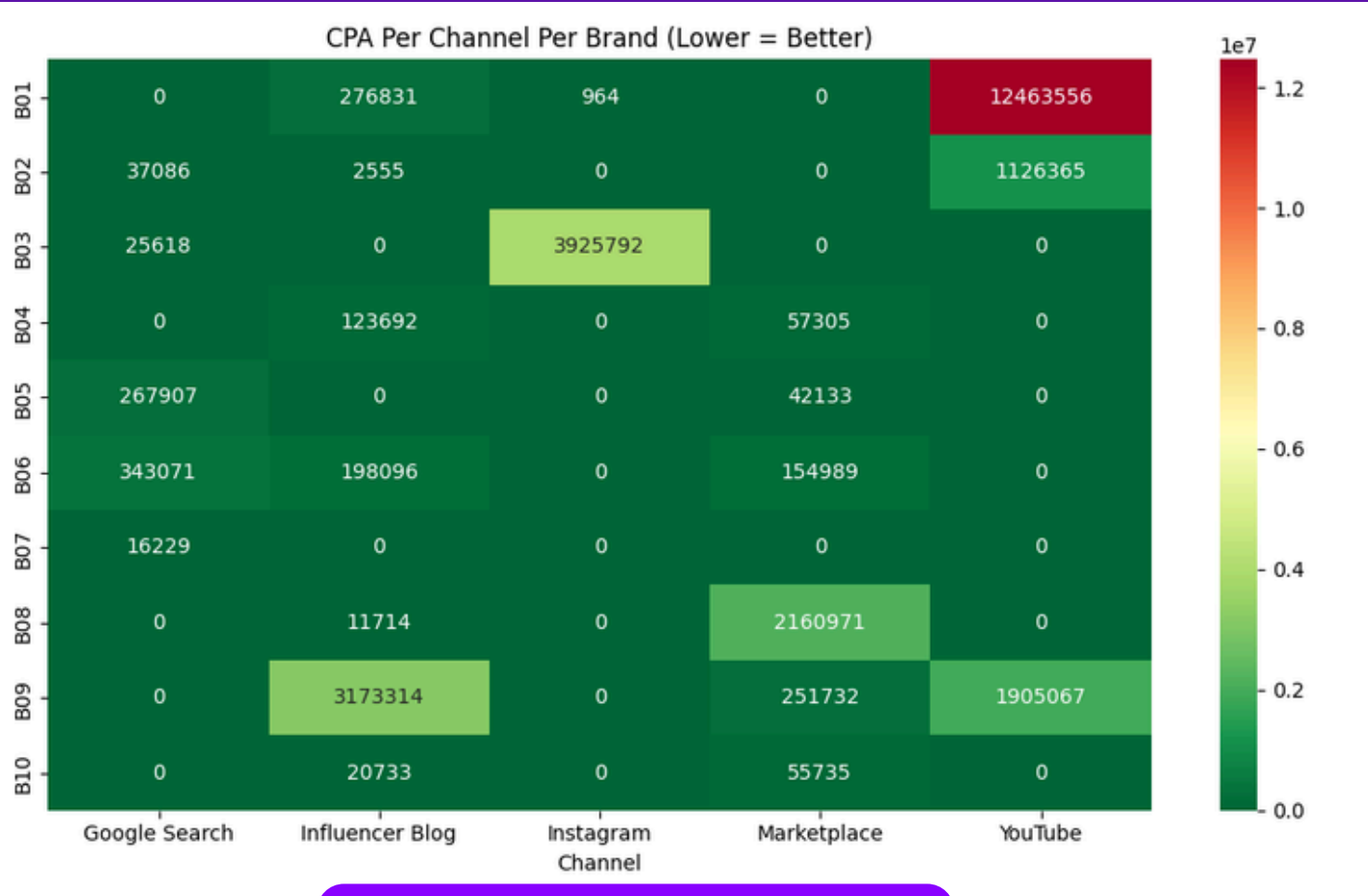
→ Most underfunded channel relative to actual performance.
- 4 No two brands have same dominant channel .

→ Proves brand-level attribution is essential.

→ Portfolio-level analysis would hide all of this.

CPA Analysis: True Cost Per Acquisition After Markov Attribution

Where Nexus Money Is Working vs Where It Is Being Wasted



What CPA Reveals

Last-Click CPA was calculated on false attribution.
It gave Marketplace 25.1% of conversions it didn't drive.
Result: Marketplace appeared efficient.
YouTube appeared to have acceptable CPA.

Markov CPA tells the truth:

- Channels with 0% attribution = infinite CPA = zero ROI.
- Budget should flow to lowest true CPA channels.
- YouTube is a budget drain across all 10 brands, 7 brands show zero attribution (infinite CPA).
- 3 brands show CPA above Rs.11 Lakh per conversion.
- Influencer Blog and Instagram are the real performers.

Key CPA Findings

1. BEST CPA IN PORTFOLIO

B01 Instagram: ₹964 per conversion
→ Most efficient channel in entire portfolio.
→ Had only ₹61L budget – criminally underfunded.
→ Should be primary investment for B01.

2. WORST CPA IN PORTFOLIO

B01 YouTube: ₹1,24,63,556 per conversion
→ Spending ₹1.24 Crore to get ONE customer.
→ 12,927x worse than Instagram for same brand.
→ Complete budget misallocation.

3. GOOGLE SEARCH EFFICIENCY

B07 Google Search: ₹16,229 per conversion.
→ Only working channel for B07.
→ Reasonable CPA but zero channel diversity.
→ Single point of failure risk.

4. INFLUENCER BLOG UNDERVALUED

B08 Influencer Blog: ₹11,714 per conversion
→ Second most efficient channel overall.
→ Last-Click was giving it only 18% credit.
→ Markov revealed it drives 94% of B08 conversions

Best vs Worst CPA

Top 5 BEST CPA

Brand	Channel	CPA	What It Means
B01	Instagram	₹964	Best in entire portfolio
B02	Influencer Blog	₹2,555	Hidden performer
B08	Influencer Blog	₹11,714	Highly efficient
B07	Google Search	₹16,229	Only working channel
B10	Influencer Blog	₹20,733	Underinvested

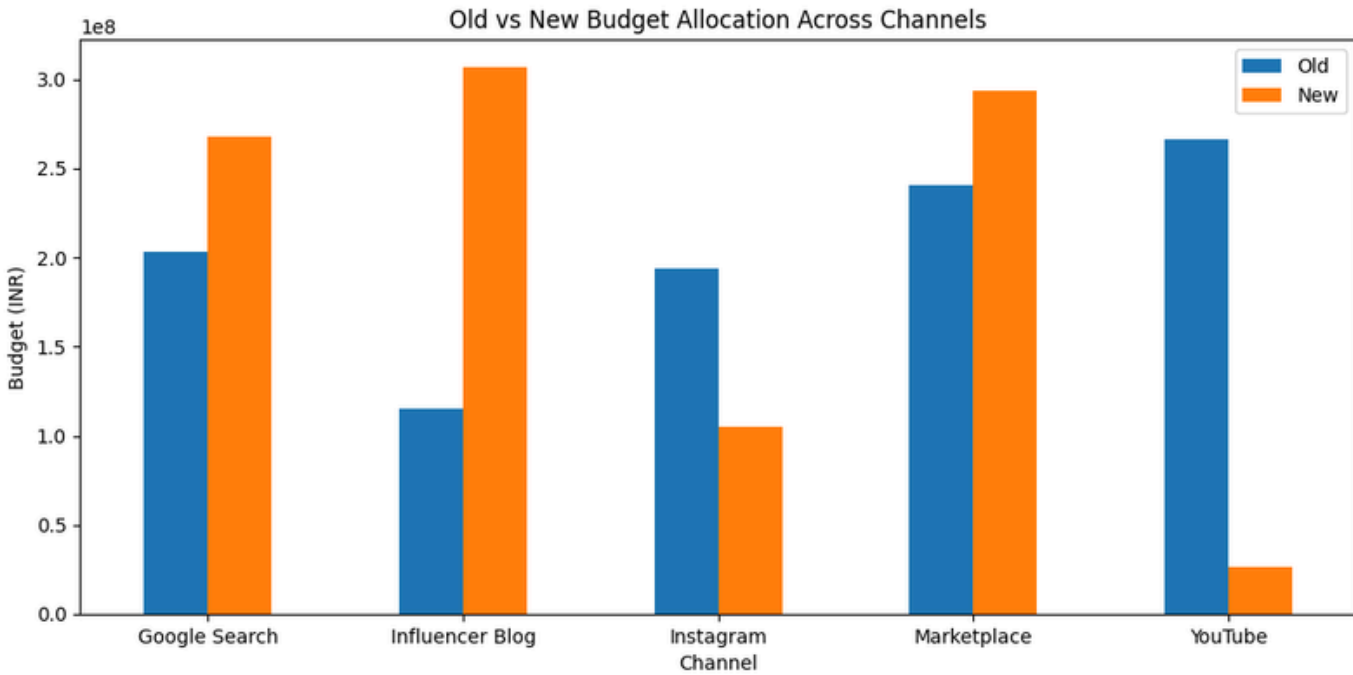
Top 5 WORST CPA

Brand	Channel	CPA	What It Means
B01	YouTube	₹1.24 Crore	Worst in portfolio
B03	Instagram	₹39.25 Lakh	Near zero attribution
B09	Influencer Blog	₹31.73 Lakh	Zero attribution
B08	Marketplace	₹21.60 Lakh	Wrong channel funded
B09	YouTube	₹19.05 Lakh	Near zero attribution

YouTube is 12,927x more expensive per conversion than Instagram FOR THE SAME BRAND

Budget Reallocation: Optimal ₹100 Crore Deployment For Next Quarter

From Gut Feel to Data-Driven Allocation | CPA-Optimized Per Brand



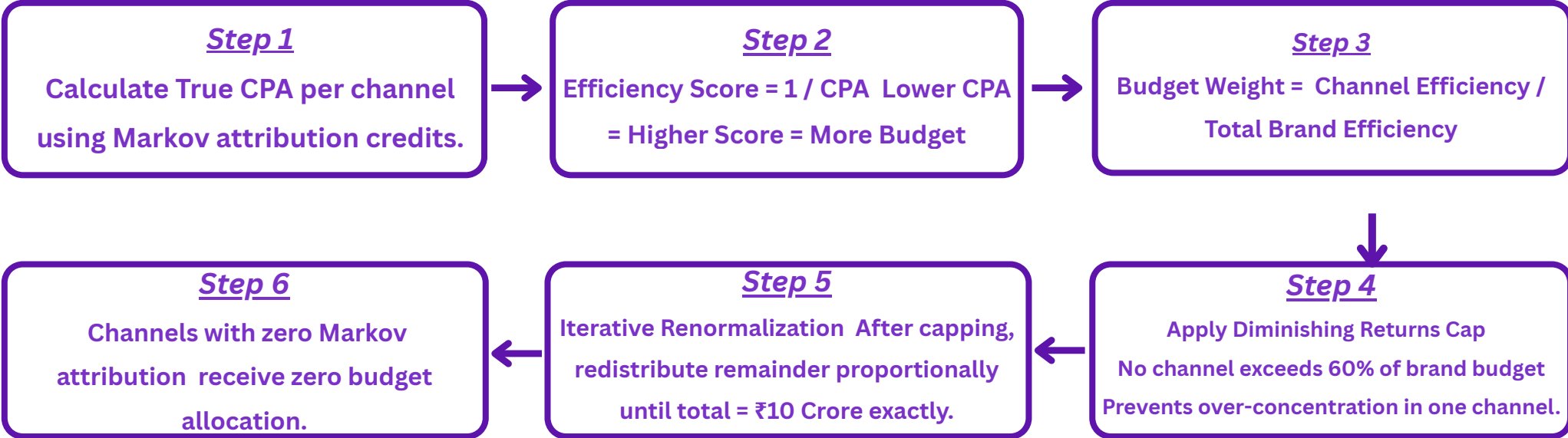
Portfolio Shift Summary

Channel	Old Budget	New Budget	Change
YouTube	₹26.60 Cr	₹2.66 Cr	-₹23.94 Cr ↓
Instagram	₹19.41 Cr	₹10.54 Cr	-₹8.87 Cr ↓
Google Search	₹20.30 Cr	₹26.81 Cr	+₹6.51 Cr ↑
Marketplace	₹24.06 Cr	₹29.32 Cr	+₹5.26 Cr ↑
Influencer Blog	₹11.49 Cr	₹30.68 Cr	+₹19.19 Cr ↑

Biggest winner: Influencer Blog +₹19.19 Cr

Biggest loser: YouTube -₹23.94 Cr

Reallocation Logic



Brand Level Reallocation Table (in Crores)

	Google Search	Influencer Blog	Instagram	Marketplace	YouTube
Brand					
B01	0.00	2.34	7.61	0.00	0.05
B02	3.76	6.12	0.00	0.00	0.12
B03	7.07	0.00	2.93	0.00	0.00
B04	0.00	3.99	0.00	6.01	0.00
B05	3.95	0.00	0.00	6.05	0.00
B06	2.02	3.50	0.00	4.48	0.00
B07	10.00	0.00	0.00	0.00	0.00
B08	0.00	7.23	0.00	2.77	0.00
B09	0.00	1.49	0.00	6.03	2.48
B10	0.00	6.01	0.00	3.99	0.00

Strategic Recommendations: 4 Actions For The CMO

From Attribution Truth to Execution – What Nexus Must Do Next

ACTION 1: DEFUND YOUTUBE IMMEDIATELY

Finding: Near-zero attribution across 9/10 brands.

Current Spend: ₹26.60 Crore.

Recommended: ₹2.66 Crore (-₹23.94 Cr).

Exceution

- Redirect ₹23.94 Cr to Influencer Blog.
- Maintain minimal YouTube presence only for B09 where it shows marginal attribution.
- Run quarterly review before any reinvestment.
- YouTube only justified if attribution improves in future quarters.

ACTION 2: SCALE INFLUENCER BLOG AGGRESSIVELY

Finding: Most underfunded channel vs performance.

Current Spend: ₹11.49 Crore.

Recommended: ₹30.68 Crore (+₹19.19 Cr).

Exceution

- B08 Influencer Blog CPA = ₹11,714 (3rd most efficient overall).
- B02 Influencer Blog CPA = ₹2,555 (2nd most efficient overall).
- Triple investment in B02 and B08 immediately.
- Influencer Blog is the hidden hero of portfolio.
- Last-Click was giving it 15% credit Markov reveals it deserves 40%+

ACTION 3: FIX B07 SINGLE CHANNEL DEPENDENCY

Finding: 100% dependent on Google Search.

Current Spend: ₹10 Cr all on Google Search.

Risk: Zero fallback if Google CPC rises.

Exceution

- Force minimum 20% budget into secondary channel testing.
- Run Instagram and Influencer Blog pilot campaigns for B07.
- Build attribution data over 2 quarters.
- Single channel dependency = existential risk for brand marketing continuity.
- Diversification is non-negotiable.

ACTION 4: IMPLEMENT QUARTERLY ATTRIBUTION AUDIT

Finding: Last-Click gave wrong signals for at least one full quarter.

Cost of Inaction: ₹23.94 Cr wasted on YouTube.

Exceution

- Run ROI Lens attribution model every quarter before budget decisions.
- Never allocate budget on Last-Click again.
- Bot detection must run before every audit.
- Attribution model should be updated as new touchpoint data becomes available.
- Build internal capability or outsource to analytics partner (Ai Palette)

Model Limitation

- **Single Quarter Data**
Model built on Q1 2026 only. Consumer behavior shifts across seasons – festive quarters produce different channel dynamics. Attribution weights derived from one quarter may not hold across the full year.
- **Synthetic Dataset**
Dataset simulates real-world conditions but is not actual Nexus data. Bot patterns and conversion rates may behave differently in production. Results must be validated against real campaign data before full budget deployment.

- **Assumed Conversion Value**
₹5,000 per conversion is an assumption. Without actual transaction revenue, ROI figures are directionally correct but not financially precise.
- **Channel Synergy Not Captured**
Markov treats each channel independently. YouTube + Instagram together may outperform either channel alone. These reinforcement effects are not modeled in the current framework.
- **Diminishing Returns Approximated**
60% cap is a proxy – not a fitted curve. True saturation modeling requires spend data across multiple budget levels over time.

Future Scope

- **Scope 1**
Multi-Quarter Integration 4+ quarters of data would capture seasonal patterns and produce more stable attribution weights per campaign period.
- **Scope 2**
Shapley Value Validation Running Shapley Value alongside Markov would cross-validate findings. Channels where both models agree can be funded with highest confidence.
- **Scope 3**
Segment Level Attribution Running separate Markov models per consumer persona would reveal that Fitness Enthusiasts and Gen-Z Trendseekers respond to completely different channel sequences.

- **Scope 4**
Real-Time Power BI Dashboard Integrating ROI Lens into Power BI would allow live campaign monitoring – triggering reallocation alerts before significant budget is wasted.
- **Scope 5**
Cross-Brand Synergy Modeling A YouTube campaign for B01 may indirectly lift awareness for B03. Modeling cross-brand effects would give Nexus a true portfolio-level channel view.

